# CURRENT ELECTRICITY – LEVEL 3 (ADVANCED MCQs / JEE MAIN PATTERN)

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## SECTION 1: NUMERICAL & CONCEPTUAL – BASIC CIRCUITS (Q1–40)

1. A conductor of resistance R, length L, and area A is bent to form a circle. The resistance:
   A. Increases  B. Decreases  C. Remains same  D. Becomes infinite
2. Wire of resistance 12 Ω is cut into 3 equal pieces, connected in parallel. New resistance:
   A. 4 Ω  B. 1.33 Ω  C. 36 Ω  D. Cannot determine
3. Two wires of same material: lengths 1:2, diameters 2:1. Ratio of resistances:
   A. 1:8  B. 8:1  C. 1:2  D. 2:1
4. Copper wire of length L, radius r, stretched keeping volume constant. New length 2L. Resistance:
   A. Doubles  B. Quadruples  C. Becomes ¼  D. Becomes ½
5. Copper wire temperature rises 0°C → 100°C, $\alpha = 0.004$ /°C. Resistance change:
   A. 40%  B. 200%  C. 400%  D. 4%
6. Battery: EMF 10 V, internal resistance 2 Ω; external R = 8 Ω. Terminal voltage:
   A. 8 V  B. 6 V  C. 10 V  D. 2 V
7. Two cells (1.5 V, 1 Ω) in series, load R = 10 Ω. Current:
   A. 0.2 A  B. 0.15 A  C. 0.3 A  D. 1 A
8. Same cells in parallel, same load. Current:
   A. 0.1 A  B. 0.15 A  C. 0.3 A  D. 1 A
9. Resistor R across battery ε, internal r. Maximum power delivered when:
   A. R = 0  B. R = r  C. R = 2r  D. R = ∞
10. Efficiency in above condition:
    A. 25%  B. 50%  C. 75%  D. 100%
11. Copper & nichrome wire, same length & area, series connection. Heat produced:
    A. More in copper  B. More in nichrome  C. Same  D. Copper none
12. Which wire becomes hotter?
    A. Copper  B. Nichrome  C. Both same  D. Depends on length
13. Drift velocity: I = 10 A, $n = 10^{28}\ m^{-3}$, $A = 10^{-6}\ m^2$. Drift velocity:
    A. $6.25 \times 10^{-4}$ m/s  B. $6.25 \times 10^{-5}$ m/s  C. $6.25 \times 10^{-6}$ m/s  D. $6.25 \times 10^{-3}$ m/s

14. If current doubled, drift velocity:
    A. Doubles B. Halves C. Remains same D. Quadruples
15. Three 20 Ω resistors: two in parallel, then series with third. Equivalent:
    A. 10 Ω B. 20 Ω C. 30 Ω D. 60 Ω
16. Wire resistance R at T; another wire, same material, R = 2R. Length ratio:
    A. 1:2 B. 2:1 C. 1:4 D. 4:1
17. Resistivity $\rho_0$ at 20°C, $\alpha = 0.004$ /°C. Resistivity at 120°C:
    A. $1.4\rho_0$ B. $1.04\rho_0$ C. $2\rho_0$ D. $0.4\rho_0$
18. Specific resistance depends on:
    A. Temperature only B. Shape/size only C. Material only D. All of above
19. Two resistors P & Q in series across V; voltage across P = 3V, Q = 2V. $R_P : R_Q$:
    A. 3:2 B. 2:3 C. 9:4 D. 4:9
20. Current through P & Q:
    A. Different B. Same C. Ratio 3:2 D. Ratio 2:3
21. Three resistors 2Ω, 3Ω, 6Ω in parallel. Equivalent resistance:
    A. 1 Ω B. 1.5 Ω C. 11 Ω D. 6 Ω
22. Voltage across combination 6 V. Total current:
    A. 3 A B. 6 A C. 9 A D. 12 A
23. Voltage across 2Ω resistor:
    A. 2 V B. 3 V C. 6 V D. 1 V
24. Current through 6Ω resistor:
    A. 1 A B. 2 A C. 3 A D. 6 A
25. Cell ε, r, connected to variable R. Maximum power dissipated in R:
    A. R = 0 B. R = r C. R = ∞ D. Always constant
26. Series circuit: R & EMF ε, power relation:
    A. $P_{total} = P_R$ always B. $P_{total} > P_R$ C. $P_{total} = P_R$ only when r = 0 D. Cannot relate
27. Wire of resistance R bent into square. Resistance between adjacent corners:
    A. R/4 B. R/2 C. 3R/4 D. R
28. Resistance between opposite corners:
    A. R/4 B. R/2 C. 3R/4 D. R
29. Circuit loop: sum of EMFs = 10 V, sum of IR drops = 10 V:
    A. Loop equation satisfied B. Violated C. Cannot determine D. Depends on direction

30. Number of Kirchhoff's equations for 2 loops, 3 branches:
    A. 1 B. 2 C. 3 D. 4
31. Meter bridge balanced when:
    A. All resistances equal B. P/Q = R/S C. Current through galvanometer D. Galvanometer shows zero
32. Potentiometer measures:
    A. Current accurately B. EMF with zero current C. Resistance D. Power directly
33. In potentiometer: $E_1$ balances at 60 cm, $E_2$ at 40 cm:
    A. $E_1 > E_2$ B. $E_1 < E_2$ C. $E_1 = E_2$ D. $E_1 = 1.5\ E_2$
34. Potentiometer wire length 100 cm, resistance 5 Ω. Internal resistance affects:
    A. Accuracy not at all B. Makes measurement impossible C. Reduces accuracy D. Increases accuracy
35. Meter bridge: unknown 8 Ω, balance at 60 cm. Unknown resistance:
    A. 12 Ω B. 6 Ω C. 4.8 Ω D. 3 Ω
36. Ammeter: shunt S, galvanometer R_g. Increase range by factor n:
    A. S = R_g/n B. S = R_g/(n-1) C. S = nR_g D. S = R_g·n
37. Voltmeter multiplier resistance M = R_g(n-1). Meaning:
    A. Voltmeter → ammeter B. Extends voltage range C. Reduces error D. Increases current capacity
38. Ideal ammeter:
    A. Infinite resistance B. Zero resistance C. Galvanometer alone D. Variable resistance
39. Ideal voltmeter:
    A. Zero resistance B. Infinite resistance C. Consumes no current D. B & C
40. High EMF circuit, r very small, short-circuit current:
    A. ε/R B. Very large C. 0 D. ε/2r

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If you want, I can **continue Section 2 (Q41–80: Complex Circuits & Network Theorems)** next, keeping the same clean advanced MCQ format for JEE Main level.

Do you want me to proceed?

yes

Perfect! Here's **SECTION 2: COMPLEX CIRCUITS & NETWORK THEOREMS (Q41–80)** in the same advanced MCQ / JEE Main style:

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## SECTION 2: COMPLEX CIRCUITS & NETWORK THEOREMS (Q41–80)

41. Thevenin equivalent resistance for a circuit seen from terminals:
    A. All sources short-circuited B. Load disconnected C. Both conditions D. Sources replaced by internal resistance
42. Norton equivalent is:
    A. Same as Thevenin B. Dual of Thevenin C. Current source with resistance D. Always for AC only
43. Superposition theorem states:
    A. Combine all effects at once B. Each source acts independently C. Total effect is algebraic sum D. Both B and C
44. Which theorem requires removing all but one source?
    A. Thevenin B. Norton C. Superposition D. Maximum power transfer
45. Reciprocity theorem applies to:
    A. Linear circuits only B. Nonlinear circuits C. AC circuits only D. Circuits with dependent sources
46. In a linear two-port network, if current $I_1$ at port 1 due to $V_2$ at port 2:
    A. $I_2 \neq I_1$ when $V_1$ and $V_2$ swapped B. $I_2 = I_1$ when conditions swapped C. No relation between currents D. Always opposite
47. Delta-Y transformation for three resistors $R_1$, $R_2$, $R_3$:
    A. Always reduces circuits B. Converts delta to Y and vice versa C. Cannot simplify circuits D. Only for parallel circuits
48. In Y-Δ transformation, Y resistors R_a, R_b, R_c:
    A. $R_1$ = R_a || R_b B. $R_1$ = (R_a·R_b + R_b·R_c + R_c·R_a)/R_c C. Cannot relate D. $R_1$ = R_a + R_b
49. Wheatstone bridge is used for:
    A. Measuring resistance accurately B. Providing maximum power C. Reducing voltage D. Rectification

50. Bridge circuit balance condition $P \cdot S = Q \cdot R$ means:
A. Null current through galvanometer  B. Maximum galvanometer current  C. $P = Q$ & $R = S$  D. All resistors equal

51. Post office box (PB) is modification of:
A. Potentiometer  B. Meter bridge  C. Wheatstone bridge  D. Ammeter

52. Slide wire bridge has:
A. Four resistances  B. Three resistances and wire  C. Only wire resistance  D. Two resistances

53. Meter bridge: unknown X opposite 5 Ω, balance at 40 cm. X =
A. 3.33 Ω  B. 2 Ω  C. 8 Ω  D. 5 Ω

54. Jockey at 50 cm (middle): unknown & known resistances:
A. Must be equal  B. Can be any value  C. Ratio 1:1  D. Product minimum

55. Sensitivity of meter bridge:
A. Highest at one end  B. Highest at center  C. Lowest at center  D. Uniform throughout

56. Loading effect in potentiometer occurs when:
A. Internal resistance = 0  B. Internal resistance high  C. Load very small  D. Cell weak

57. To minimize loading effect in potentiometer:
A. Increase internal resistance  B. Decrease potentiometer resistance  C. Use cell with low internal resistance  D. Increase load resistance

58. Galvanometer $R_g = 50\ \Omega$, $I_g = 1$ mA. To measure 10 A, shunt needed:
A. 0.5 Ω  B. 0.05 Ω  C. 5 Ω  D. 50 Ω

59. Multiplying factor for above:
A. 100  B. 1000  C. 10000  D. 50000

60. Galvanometer to measure 10 V: $R_g = 50\ \Omega$, $I_g = 1$ mA. Multiplier resistance:
A. 9950 Ω  B. 950 Ω  C. 9999 Ω  D. 10000 Ω

61. Two circuits are equivalent if:
A. Same components  B. Same V-I characteristics at terminals  C. Same power dissipation  D. All of above

62. Balanced Wheatstone bridge: galvanometer reads:
A. Shows current  B. Shows voltage  C. Reads zero  D. Oscillates

63. Unbalanced bridge, potentials at four corners $V_A$, $V_B$, $V_C$, $V_D$:
A. $V_A = V_B$  B. $V_A = V_C$  C. $V_A = V_D$  D. $V_A \neq V_B$

64. Balanced condition can also be written as:

A. $Z_1/Z_3 = Z_2/Z_4$  B. $Z_1 \cdot Z_4 = Z_2 \cdot Z_3$  C. Both A & B  D. $Z_1 + Z_3 = Z_2 + Z_4$

65. AC bridge differs from DC because:

A. Uses complex impedances  B. Frequency dependent  C. Phase angle matters  D. All of above

66. Maximum power theorem: P_max occurs when:

A. R_L = 0  B. R_L = R_th  C. R_L = ∞  D. R_L = 2R_th

67. Efficiency at maximum power transfer:

A. 0%  B. 50%  C. 75%  D. 100%

68. Source resistance >> load: power transfer is:

A. Efficient but small  B. Inefficient but large  C. Maximum  D. Cannot determine

69. Adaptive impedance matching:

A. Fixes load at source resistance  B. Adjusts load for max power  C. Reduces voltage  D. Increases current only

70. Non-linear circuit:

A. Superposition applies  B. Superposition doesn't apply  C. Thevenin applies  D. Kirchhoff fails

71. Dependent sources (VCVS, VCCS):

A. Follow superposition strictly  B. Require special handling  C. Always in AC circuits  D. Cannot analyze

72. Nodal analysis advantage:

A. Fewer equations than loop  B. Direct voltage calculation  C. Handles voltage sources easily  D. All of above

73. Mesh analysis advantage:

A. Fewer equations for current sources  B. Direct current calculation  C. Loop currents easier to track  D. All of above

74. Circuit with n nodes, independent node voltage equations:

A. n  B. n-1  C. n+1  D. 2n

75. Circuit with b branches & n nodes, independent loop equations:

A. b - n  B. b - n + 1  C. n - 1  D. b - 1

76. First law of Kirchhoff also called:

A. Voltage law  B. Current law  C. Power law  D. Energy law

77. Second law of Kirchhoff also called:

A. Junction rule  B. Loop rule  C. Mesh rule  D. Node rule

78. Circuit with time-varying fields, Kirchhoff's voltage law:
A. Holds exactly  B. Not valid  C. Needs modification  D. Fails

79. Sign convention for element: current enters positive terminal:
A. Element supplies energy  B. Element absorbs energy  C. No power transfer  D. Depends on resistance

80. Power conservation in closed loop:
A. ΣP_supplied = ΣP_dissipated  B. ΣP varies with time  C. Total power always zero  D. Cannot be conserved

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That completes **Q41–80 (SECTION 2)**.

I can now continue with **SECTION 3: ENERGY, POWER & HEATING (Q81–110)** in the same detailed JEE-style format.

Do you want me to proceed with Section 3 next?

yes

Great! Here's **SECTION 3: ENERGY, POWER & HEATING (Q81–110)** in advanced JEE Main MCQ style:

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## SECTION 3: ENERGY, POWER & HEATING (Q81–110)

81. Joule's heating formula $H = I^2Rt$ means:
A. Heat independent of I  B. Heat depends on square of I  C. Heat proportional to R  D. Both B and C

82. If current doubles, heat produced becomes:
A. Double  B. Four times  C. Half  D. Quarter

83. Heating effect depends on:
A. Current direction  B. Resistance value  C. Time duration  D. Both B and C

84. In a conductor, heat is generated due to:
A. Random motion of electrons  B. Collision with atoms  C. Resistance opposing flow  D. All of above

85. Power delivered to external resistor R with EMF ε and internal r:

A. $\frac{\varepsilon^2 R}{(R+r)^2}$ B. $\frac{\varepsilon^2 R}{R+r}$ C. $\frac{\varepsilon^2}{(R+r)^2}$ D. $\varepsilon I$

86. Power dissipated in internal resistance:

A. $I^2 r$ B. $\varepsilon I - I^2 R$ C. $V^2/r$ D. Both A and B

87. Total power from cell:

A. $\varepsilon I$ B. $VI$ C. $I^2(R+r)$ D. All of above

88. Electrical energy consumed in time t with resistance R:

A. $I^2 Rt$ B. $V^2 t/R$ C. $VIt$ D. All of above

89. 100 W bulb operates at 10 V:

A. Current = 10 A B. Resistance = 1 Ω C. Resistance = 10 Ω D. Cannot determine

90. Cost of electricity depends on:

A. Power only B. Time only C. Power and time D. Cost fixed

91. Two resistors in series, same current, heat ratio:

A. $R_1 : R_2$ B. $R_2 : R_1$ C. 1:1 D. Indeterminate

92. Two resistors in parallel, same voltage, heat ratio:

A. $R_1 : R_2$ B. 1:1 C. $1/R_1 : 1/R_2$ D. $R_2 : R_1$

93. Star fuse rating depends on:

A. Current capacity B. Heat rating C. Voltage rating D. All of above

94. Fuse melts when:

A. Current exceeds rating B. Temperature exceeds melting point C. Resistance becomes zero D. Both A and B

95. Circuit breaker preferred over fuse because:

A. Reusable B. No replacement C. Can be reset D. All of above

96. Earth leakage circuit breaker protects against:

A. Overcurrent B. Short circuit C. Electric shock D. All of above

97. In a circuit, if $P = VI$ and $V = IR$:

A. $P = I^2 R$ B. $P = V^2/R$ C. Both A and B D. Neither A nor B

98. Temperature rise in conductor depends on:

A. Heat generated only B. Heat dissipation rate C. Both D. Material

99. Specific heat capacity affects:

A. Heat generated B. Temperature rise C. Current flow D. Resistance

100. Thermal equilibrium reached when:

A. No more heat generated B. Heat generation = Heat loss C. Temperature constant D. Both B and C

101. Superconductor below critical temperature:
    A. High resistance  B. Zero resistance  C. Becomes insulator  D. Generates heat
102. Persistent current in superconductor:
    A. Decays slowly  B. Decays exponentially  C. Does not decay  D. Oscillates
103. Superconductivity disappears when:
    A. Temperature above critical  B. Current above critical  C. Magnetic field above critical  D. All of above
104. BCS theory explains superconductivity through:
    A. Electron-phonon interaction  B. Electron pairing  C. Cooper pairs  D. All of above
105. Meissner effect means:
    A. Magnetic field enters superconductor  B. Magnetic field expelled  C. No magnetic field inside  D. Both B and C
106. Seebeck effect produces:
    A. Current from temperature difference  B. Heat from current  C. Cooling from current  D. None
107. Peltier effect application:
    A. Thermocouple  B. Refrigeration  C. Heating  D. All of above
108. Thomson effect is:
    A. Heat absorption in current-carrying conductor with temperature gradient  B. Magnetic field generation  C. Only in semiconductors  D. Opposite of Seebeck
109. Figure of merit for thermoelectric device:
    A. $Z = S^2\sigma/\kappa$  B. $Z = \sigma/\kappa$  C. $Z = S/\kappa$  D. $Z = \sigma S$
110. Hall effect principle used for:
    A. Determining charge carrier type  B. Measuring magnetic field  C. Finding carrier concentration  D. All of above

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That completes **Q81–110 (SECTION 3)**. ✅

Next, I can continue with **SECTION 4: HIGH-LEVEL CONCEPTUAL PROBLEMS (Q111–130)** to finish the full 130-question set.

Do you want me to go ahead with Section 4?

yes

Perfect! Here's **SECTION 4: HIGH-LEVEL CONCEPTUAL PROBLEMS (Q111–130)** in advanced JEE Main MCQ style:

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## SECTION 4: HIGH-LEVEL CONCEPTUAL PROBLEMS (Q111–130)

111. Resistance of a rectangular conductor with length L, width W, height H, resistivity ρ:
A. $\rho L/(W \times H)$ B. $\rho W/(L \times H)$ C. $\rho H/(L \times W)$ D. Cannot determine
112. If resistivity varies as $\rho = \rho_0(1 + \alpha T)$, resistance temperature variation:
A. Linear only B. Non-linear C. Parabolic D. Exponential
113. In a tapered wire, current density:
A. Same throughout B. Higher in thin part C. Higher in thick part D. Variable unpredictably
114. Current distribution in parallel paths:
A. Always equal B. Inversely proportional to resistance C. Proportional to resistance D. Random
115. Skin effect in AC conductors means:
A. Current uniform throughout B. Current at surface only C. Current at center only D. Current distributed by frequency
116. Transmission losses in power lines depend on:
A. Current only B. Resistance only C. $I^2R$ D. Voltage rating
117. Step-up transformer reduces losses because:
A. Increases power B. Decreases current (for same power) C. Decreases resistance D. Increases voltage
118. Two identical cells in series vs parallel:
A. Series: higher EMF and current B. Parallel: higher current C. Series: higher terminal voltage D. Both A and C
119. For maximum current from battery, load resistance should be:
A. Zero B. Very high C. Equal to internal resistance D. Twice internal
120. For maximum power transfer, efficiency sacrifice:
A. 25% lost internally B. 50% lost internally C. 75% efficiency only D. No loss
121. In bridge circuit at balance, power dissipation in galvanometer:
A. Maximum B. Minimum (zero) C. Moderate D. Oscillating

122. Unknown resistance measurement by potentiometer:

A. Requires zero current B. Requires maximum current C. Requires specific current D. Current irrelevant

123. Null point method advantage:

A. No galvanometer calibration needed B. High accuracy C. Small EMF measurable D. All of above

124. In multi-cell battery, internal resistance:

A. Always adds up B. Adds in series connection C. Adds in parallel (different rule) D. Both A and B for series

125. Lenz's law in circuit context:

A. Opposes EMF changes B. EMF opposes current change C. Current opposes voltage change D. All equivalent

126. Self-inductance opposes:

A. Current increase B. Current decrease C. Both D. Neither

127. Mutual inductance between coils:

A. Always positive B. Can be negative or positive C. Zero always D. Depends on direction only

128. AC circuit with pure resistance:

A. Current and voltage in phase B. Current leads voltage C. Voltage leads current D. 90° phase difference

129. AC circuit with pure inductance:

A. Current and voltage in phase B. Current lags voltage by 90° C. Current leads voltage by 90° D. Variable phase

130. In resonant circuit at resonance:

A. Impedance maximum B. Impedance minimum C. Impedance equals resistance D. Both B and C